

ToyModelSOEMC: Equation Summary and Symbol Glossary

From ToyModelSOEMC.mod

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1 Scope

This note summarizes the full model equation system in `ToyModelSOEMC.mod`, including commodity-level dynamics, aggregation, equilibrium conditions, and shock processes. The commodity set in this training file is $\mathcal{C} = \{\text{oil, coal}\}$, but equations are written in generic form for any $c \in \mathcal{C}$.

2 Index Sets and Conventions

- Time is discrete: $t = 0, 1, 2, \dots$
- Commodity index: $c \in \mathcal{C}$, with $|\mathcal{C}| = N_{\text{COMM}}$.
- Dynare timing: $x(-1) = x_{t-1}$, $x(+1) = x_{t+1}$.
- $\max(\cdot, \cdot)$ and $\min(\cdot, \cdot)$ are pointwise nonlinear operators.

3 Per-Commodity Equations (for each $c \in \mathcal{C}$)

1) Price shock state (AR(1))

$$\varepsilon_{c,t} = \rho_p \varepsilon_{c,t-1} + \nu_{c,t}. \quad (1)$$

Symbol	Meaning
$\varepsilon_{c,t}$	Price shock state for commodity c at time t .
ρ_p	Persistence parameter of the AR(1) shock process.
$\varepsilon_{c,t-1}$	Lagged price shock state.
$\nu_{c,t}$	Innovation to the price shock state.
c	Commodity index in $\mathcal{C}$.
t	Time index.

2) Reserve drawdown rule

$$R_{c,t}^{\text{draw}} = \min\left(\phi_R \max(0, -e_{c,t}^{\text{shock}}), R_{c,t-1}\right). \quad (2)$$

Symbol	Meaning
$R_{c,t}^{\text{draw}}$	Reserve release of commodity c in period t .
ϕ_R	Policy parameter for desired release fraction of shortage.
$e_{c,t}^{\text{shock}}$	Supply shock to imports; negative values imply shortages.
$R_{c,t-1}$	Available reserve stock at beginning of period.
$\max(0, -e_{c,t}^{\text{shock}})$	Positive shortage magnitude.
$\min(\cdot, \cdot)$	Caps release by available reserves.

3) Imported dirty energy flow

$$E_{D,c,t}^{\text{import}} = \bar{E}_{D,c} + e_{c,t}^{\text{shock}}. \quad (3)$$

Symbol	Meaning
$E_{D,c,t}^{\text{import}}$	Imported dirty energy of commodity c .
$\bar{E}_{D,c}$	Baseline import level of commodity c .
$e_{c,t}^{\text{shock}}$	Supply disruption shock to imports.

4) Total dirty energy use of commodity c

$$E_{D,c,t} = E_{D,c,t}^{\text{import}} + R_{c,t}^{\text{draw}}. \quad (4)$$

Symbol	Meaning
$E_{D,c,t}$	Total dirty energy used from commodity c .
$E_{D,c,t}^{\text{import}}$	Imported dirty energy flow.
$R_{c,t}^{\text{draw}}$	Reserve release flow.

5) Energy services from commodity c

$$ES_{c,t} = E_{D,c,t} \left(\frac{Q_{c,t-1}}{\bar{Q}_c} \right)^{1-\alpha_Q}. \quad (5)$$

Symbol	Meaning
$ES_{c,t}$	Useful energy services from commodity c .
$E_{D,c,t}$	Total dirty energy flow for commodity c .
$Q_{c,t-1}$	Installed conversion capacity at start of period.
$\bar{Q}_c$	Capacity normalization benchmark.
α_Q	Raw energy share parameter in service production.
$1 - \alpha_Q$	Capacity elasticity in the service multiplier.

6) Capacity investment rule

$$IQ_{c,t} = \phi_{IQ} \frac{Y_t}{N_{\text{COMM}}}. \quad (6)$$

Symbol	Meaning
$IQ_{c,t}$	Capacity investment for commodity c .
ϕ_{IQ}	Output share allocated to total capacity investment.
Y_t	Output (GDP).
N_{COMM}	Number of commodities; investment is split equally.

7) Capacity accumulation

$$Q_{c,t} = (1 - \delta_Q)Q_{c,t-1} + IQ_{c,t}. \quad (7)$$

Symbol	Meaning
$Q_{c,t}$	End-of-period installed capacity.
δ_Q	Depreciation rate of installed capacity.
$Q_{c,t-1}$	Beginning-of-period capacity stock.
$IQ_{c,t}$	Capacity investment flow.

8) Reserve stock dynamics

$$R_{c,t} = (1 - \delta_R)R_{c,t-1} - R_{c,t}^{\text{draw}} + \delta_R \bar{R}_c + \gamma_R \max(0, \bar{R}_c - R_{c,t-1}). \quad (8)$$

Symbol	Meaning
$R_{c,t}$	End-of-period reserve stock.
δ_R	Physical loss rate of reserves.
$R_{c,t-1}$	Beginning-of-period reserve stock.
$R_{c,t}^{\text{draw}}$	Reserve release in period t .
$\bar{R}_c$	Target/benchmark reserve stock.
γ_R	Refill speed toward target after drawdown.
$\max(0, \bar{R}_c - R_{c,t-1})$	Positive reserve gap to be refilled.

9) Terms-of-trade cost by commodity

$$ToT_{c,t} = \varepsilon_{c,t} E_{D,c,t}^{\text{import}}. \quad (9)$$

Symbol	Meaning
$ToT_{c,t}$	Terms-of-trade cost for commodity c .
$\varepsilon_{c,t}$	Price shock state for commodity c .
$E_{D,c,t}^{\text{import}}$	Imported dirty energy flow.

4 Aggregation and Energy Block

10) Total dirty-energy services

$$E_{D,t}^{\text{service}} = \sum_{c \in \mathcal{C}} ES_{c,t}. \quad (10)$$

Symbol	Meaning
$E_{D,t}^{\text{service}}$	Aggregate dirty-energy services.
$\sum_{c \in \mathcal{C}}$	Sum across all commodities.
$ES_{c,t}$	Commodity-level energy services.
$\mathcal{C}$	Set of dirty-energy commodities.

11) Total energy input to production

$$E_t = E_{D,t}^{\text{service}} + \bar{E}_C. \quad (11)$$

Symbol	Meaning
E_t	Total energy input used in production.
$E_{D,t}^{\text{service}}$	Dirty-energy services aggregate.
$\bar{E}_C$	Exogenous clean-energy baseline.

5 Production and Factor Prices

Define

$$\Xi_t = \alpha_K K_{t-1}^\rho + \alpha_L \bar{L}^\rho + \alpha_E E_t^\rho. \quad (12)$$

Symbol	Meaning
Ξ_t	Inner CES aggregator.
$\alpha_K, \alpha_L, \alpha_E$	CES share parameters for capital, labor, and energy.
K_{t-1}	Predetermined capital stock.
$\bar{L}$	Exogenous labor supply level.
E_t	Total energy input.
ρ	CES curvature parameter.

12) Output production (CES)

$$Y_t = A \Xi_t^{1/\rho}. \quad (13)$$

Symbol	Meaning
Y_t	Output (GDP).
A	Total factor productivity scale parameter.
Ξ_t	CES inner aggregate of inputs.
ρ	CES curvature parameter.

13) Rental rate of capital

$$r_t = A \alpha_K K_{t-1}^{\rho-1} \Xi_t^{1/\rho-1}. \quad (14)$$

Symbol	Meaning
r_t	Rental rate of capital.
A	Total factor productivity scale parameter.
α_K	Capital share parameter in CES production.
K_{t-1}	Predetermined capital stock.
ρ	CES curvature parameter.
Ξ_t	CES inner aggregate of inputs.

14) Wage rate

$$w_t = A \alpha_L \bar{L}^{\rho-1} \Xi_t^{1/\rho-1}. \quad (15)$$

Symbol	Meaning
w_t	Wage rate.
A	Total factor productivity scale parameter.
α_L	Labor share parameter in CES production.
$\bar{L}$	Exogenous labor supply level.
ρ	CES curvature parameter.
Ξ_t	CES inner aggregate of inputs.

15) Energy shadow price

$$p_{E,t} = A \alpha_E E_t^{\rho-1} \Xi_t^{1/\rho-1}. \quad (16)$$

Symbol	Meaning
$p_{E,t}$	Shadow price/marginal product of energy input.
A	Total factor productivity scale parameter.
α_E	Energy share parameter in CES production.
E_t	Total energy input.
ρ	CES curvature parameter.
Ξ_t	CES inner aggregate of inputs.

6 Resource Constraint and Intertemporal Conditions

16) Aggregate resource constraint

$$Y_t = C_t + I_t + \sum_{c \in \mathcal{C}} \left[IQ_{c,t} + ToT_{c,t} + \kappa_{1,c} R_{c,t-1} + \kappa_{2,c} R_{c,t-1}^2 \right]. \quad (17)$$

Symbol	Meaning
Y_t	Output available as resources in period t .
C_t	Consumption.
I_t	Physical capital investment.
$\sum_{c \in \mathcal{C}}$	Sum of commodity-specific costs and investment.
$IQ_{c,t}$	Capacity investment for commodity c .
$ToT_{c,t}$	Terms-of-trade import cost for commodity c .
$\kappa_{1,c} R_{c,t-1}$	Linear reserve carrying cost.
$\kappa_{2,c} R_{c,t-1}^2$	Quadratic reserve infrastructure/storage cost.
$\kappa_{1,c}, \kappa_{2,c}$	Cost coefficients by commodity.

17) Capital accumulation

$$K_t = (1 - \delta)K_{t-1} + I_t. \quad (18)$$

Symbol	Meaning
K_t	End-of-period physical capital stock.
δ	Depreciation rate of physical capital.
K_{t-1}	Beginning-of-period capital stock.
I_t	Investment flow.

18) Euler equation

$$\frac{1}{C_t} = \beta \frac{1}{C_{t+1}} (r_{t+1} + 1 - \delta). \quad (19)$$

Symbol	Meaning
C_t	Current consumption.
C_{t+1}	Next-period consumption.
β	Intertemporal discount factor.
r_{t+1}	Next-period rental return on capital.
δ	Capital depreciation rate.
$1/C_t$	Marginal utility under log utility.

19) Net exports definition

$$NX_t = - \sum_{c \in \mathcal{C}} ToT_{c,t}. \quad (20)$$

Symbol	Meaning
NX_t	Net exports.
$\sum_{c \in \mathcal{C}}$	Sum across all commodities.
$ToT_{c,t}$	Terms-of-trade cost by commodity.
–	Sign convention: positive ToT costs reduce net exports.

7 Shock Specifications in the .mod File

20) Zero-mean initialization for iid shocks

$$\nu_{c,t} = 0, \quad e_{c,t}^{\text{shock}} = 0 \quad \text{in initval.} \quad (21)$$

Symbol	Meaning
$\nu_{c,t}$	Price innovation shock for commodity c .
$e_{c,t}^{\text{shock}}$	Supply disruption shock for commodity c .
$= 0$	Initialization value in the deterministic baseline setup.
initval	Dynare initialization block for starting values.

21) Standard deviations used in shocks block

For the training file values:

$$\nu_{\text{oil},t} \sim \mathcal{N}(0, 0.20^2), \quad e_{\text{oil},t}^{\text{shock}} \sim \mathcal{N}(0, 0.07^2), \quad (22)$$

$$\nu_{\text{coal},t} \sim \mathcal{N}(0, 0.22^2), \quad e_{\text{coal},t}^{\text{shock}} \sim \mathcal{N}(0, 0.07^2). \quad (23)$$

Symbol	Meaning
$\sim \mathcal{N}(\mu, \sigma^2)$	Normally distributed random variable with mean μ and variance σ^2 .
$\nu_{\text{oil},t}, \nu_{\text{coal},t}$	Price innovation shocks for oil and coal.
$e_{\text{oil},t}^{\text{shock}}, e_{\text{coal},t}^{\text{shock}}$	Supply disruption shocks for oil and coal.
0.20, 0.22	Standard deviations of price innovation shocks.
0.07	Standard deviation of supply disruption shocks in EP mode calibration.

8 Parameter Identities Explicitly Defined in the File

The model file also includes calibration equations/identities:

$$\rho = \frac{\eta - 1}{\eta}, \quad (24)$$

Symbol	Meaning
ρ	CES curvature parameter used in production and factor prices.
η	Elasticity of substitution in the CES specification.
$\rho = (\eta - 1)/\eta$	Algebraic mapping from elasticity to Dynare curvature form.

$$\kappa_{1,\text{oil}} = \delta_R \bar{p}_D, \quad \kappa_{1,\text{coal}} = \delta_R \bar{p}_D, \quad (25)$$

Symbol	Meaning
$\kappa_{1,\text{oil}}, \kappa_{1,\text{coal}}$	Linear reserve carrying-cost coefficients.
δ_R	Physical reserve loss rate.
$\bar{p}_D$	Baseline dirty-energy world price.

$$\bar{R}_{\text{oil}} = \frac{90}{365} \bar{E}_{D,\text{oil}}, \quad \bar{R}_{\text{coal}} = \frac{90}{365} \bar{E}_{D,\text{coal}}. \quad (26)$$

Symbol	Meaning
$\bar{R}_{\text{oil}}, \bar{R}_{\text{coal}}$	Reserve targets for oil and coal.
$90/365$	Conversion of 90-day coverage to annual model units.
$\bar{E}_{D,\text{oil}}, \bar{E}_{D,\text{coal}}$	Baseline annual dirty-energy imports for oil and coal.

9 Notes

- The .mod file uses a macro loop to instantiate equations for each commodity. The equations above are exact generic forms of those generated blocks.
- For the current configuration, substitute $c \in \{\text{oil}, \text{coal}\}$ to obtain all explicit equation instances.